Jaya Sharma

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Introduction

I'm a PhD student in Cybersecurity Engineering at the Technical University of Denmark (DTU), working under the supervision of Dr. Christian Majenz. My current research lies at the intersection of theoretical cryptography and quantum computing, with a focus on the security of cryptographic constructions in the quantum random oracle model. My other specific interests include public key cryptography, zero-knowledge proofs, lattice-based cryptography and quantum information.

Education

Technical University of Denmark (DTU) 2024-27
PhD in Cybersecurity Engineering, ***Supervisor:*** Dr. Christian Majenz

Indraprastha Institute of Information Technology Delhi (IIITD) 2022-24
PhD Junior Research Fellow in Computer Science and Engineering *(completed 3 semesters)*

Indian Institute of Technology Roorkee (IITR) 2020-22
Master of Science in Mathematics Grade: 8.675/10

University of Delhi- Lady Shri Ram College For Women (LSR) 2017-20
Bachelor of Science (Honors) in Mathematics Grade: 9.081/10

Research Visits

Centrum Wiskunde & Informatica (CWI) 09/2026-01/2027
Host: Dr. Serge Fehr

Internships

QWorld Association-QIntern *(Remote internship)* 07-09/2025
Project: Implementing Efficient block-encoding of dense matrices via QRAM.

Indian Institute of Science, Bangalore (IISc) 06-07/2023
Mentor: Dr. Chaya Ganesh, ***Project:*** Compression theory of sigma protocols and lattice based sigma protocols.

Indian Institute of Science Education and Research (IISER) Bhopal*(Remote internship)* 06-08/2021
Mentor: Dr. Sanjay Singh, ***Project:***Introduction to Algebraic Function Fields and Codes

Publications/Projects

Security of the Fischlin's Transform in Quantum Random Oracle Model 2026
QCrypt'26, ASIACRYPT'26
Christian Majenz, Jaya Sharma

Implementating Efficient block-encoding of dense matrices via QRAM using the Qrisp framework 2025
QWorld QIntern Project 2025(Library currently in review by Qrisp Developers)
Viraj Dsouza, Kalpesh Bhatnagar, Kyle Carlo Lasala, Jaya Sharma

A Survey on Quantum Error Correcting Codes 2023

Elliptic Curves Over Complex Numbers- Master's Thesis 2022
Supervisor- Dr. A. Swaminathan

Brent-Dekker Method: An Attempt to Decode 2020
Eclat, A Mathematics Journal, Vol 11
Dr. Monika Singh, Shagun Agrawal, Jaya Sharma

Presentations

Security of the Fischlin Transform in Quantum Random Oracle Model- QCrypt 08/2026
Fiat Shamir Transform in QROM- Crypto Reading Group DTU 10/2024
Lattice Based Sigma Protocols- Theory Reading Group IIITD 10/2023
Trapdoors for Hard Lattices and New Crytographic Constructions-Theory Reading Group IIITD 05/2023
The Rabin Cryptosystem: Mathematical Foundations and Implementation- IIITD 12/2022
Coherent Measures of Financial Risk- MSc Mathematics Seminar, IITR 10/2021
Elliptic Curve Cryptography- Department of Mathematics, LSR 05/2019
Latin Squares- Recreation, Disproving Euler to Cryptography- Department of Mathematics, LSR 05/2018
Applications of Congruence Modulo- Department of Mathematics LSR 02/2018

Teaching Assistance

Discrete Mathematics (DTU) 2024,2025
Graph Theory (DTU) 2025
Applied Cryptography (DTU) 2024
Introduction To Cryptography (IIT Delhi) 2023
Discrete Mathematics (IIITD) 2022

Honours

Thomas B. Thriges Fond Grant 2026
William Demat Fonden Grant 2026
Otto Mønsteds Fond Grant 2026
-Grants to fund research stay at CWI Amsterdam.

ACM-India Anveshan Setu Fellowship 2023
- Awarded by Association for Computing Machinery(ACM) India to work at IISc.

Joint CSIR-UGC NET (JRF) Fellowship Mathematical Sciences 2022
-Awarded PhD Research Fellowship for securing All India Rank 122/30942 candidates in the exam conducted jointly by Council for Scientific and Industrial Research and University Grants Commission India.

IIT JAM Mathematics: secured All India Rank 143/14374 candidates 2020
- Joint Admission Test for Master of Science Courses at Centrally Funded Technical Institutions in India.

Useful Courses Completed

Introduction to Cryptography, Lattices in Computer Science, Quantum Mechanics, Introduction to Quantum Computing with Qiskit, Advanced Linear Algebra, Abstract Algebra(Group,Ring and Field Theory), Operator Theory, Elementary Number Theory, Proabability and Statistics, Real Analysis, Complex Analysis, Numerical Analysis, Financial Mathematics, Calculus, Ordinary and Partial Differential Equations, C++ Programming, Python.

Workshops/Summer Schools/Conferences

Qcrypt- Ottawa, Canada 08/2026
Eurocrypt- Rome, Italy 05/2026
DQC Scientific Quantum Conference 2026- Aalborg University, Denmark 04/2026
Theory of Cryptography Conference (TCC) 2025-Aarhus, Denmark 12/2025
Workshop on Mathematics of Post-Quantum Cryptography- University of Zurich 06/2025
Quantum Safe Internet (QSI) Workshop- Technical University of Denmark, Lyngby 05/2025
IACR Summer School on Post Quantum Cryptography- University of Warsaw 07/2024
ACM India Grad Cohort- IISc Bangalore 07/2023
CSE Research Symposium- IIT Bombay 03/2023
Vigyan Vidushi Mathematics Summer School- TIFR Mumbai 07/2021
STCS Vigyan Vidushi Summer School on Theoretical Computer Science- TIFR Mumbai 06/2021
Workshop: Calculus the Geometrical Way- Miranda House College, University of Delhi 09/2019

Other Activities

PhD Committee Member- DTU Compute 2025
Illustrator-The LSR College Magazine 2018-19

Languages

English, Danish (completed DU3, A2 level), Hindi (mother tongue).